

HW 4.2.0: Angles

1. Indicate each angle on a circle: 60° , 280° , -225° , 10° , $\frac{7\pi}{6}$, $\frac{9\pi}{4}$
2. Indicate each angle on a circle: 120° , 345° , -45° , 220° , $\frac{4\pi}{3}$, $\frac{5\pi}{4}$
3. Convert the angle 90° to radians.
4. Convert the angle 60° to radians.
5. Convert the angle $\frac{2\pi}{3}$ from radians to degrees.
6. Convert the angle $\frac{7\pi}{6}$ from radians to degrees.
7. Convert the angle $\frac{11\pi}{12}$ from radians to degrees.
8. Convert the angle $-\frac{3\pi}{15}$ from radians to degrees.
9. Convert the angle 280° from degrees to radians.
10. Convert the angle -288° from degrees to radians.
11. Find the angle between 0° and 360° that is coterminal with a 645° angle.

12. Find the angle between 0° and 360° that is coterminal with a 443° angle.

13. Find the angle between 0° and 360° that is coterminal with a -1682° angle.

14. Find the angle between 0° and 360° that is coterminal with a -1300° angle.

15. Find the angle between 0 and 2π in radians that is coterminal with the angle $\frac{23\pi}{9}$.

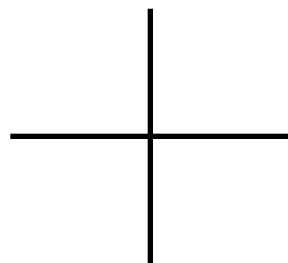
16. Find the angle between 0 and 2π in radians that is coterminal with the angle $\frac{23\pi}{3}$.

17. Find the angle between 0 and 2π in radians that is coterminal with the angle $-\frac{7\pi}{2}$.

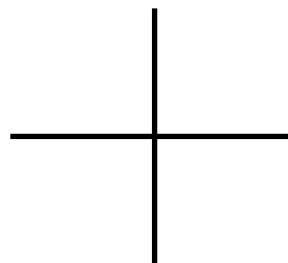
18. Find the angle between 0 and 2π in radians that is coterminal with the angle $-\frac{25\pi}{6}$.

Given a trigonometric function value and the specified quadrant, find the other five trigonometric function values. (Hint: draw a triangle in the quadrant and think about the signs of x and y)

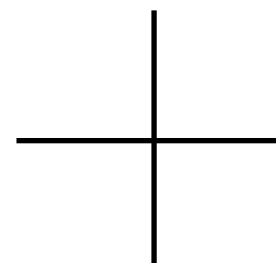
19. $\sin(A) = \frac{5}{13}$ where A is in Quadrant I



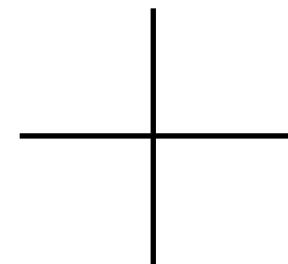
20. $\sec(A) = -\frac{5}{3}$ where A is in Quadrant II



21. $\csc(A) = -\frac{25}{24}$ where $\pi \leq A \leq \frac{3\pi}{2}$

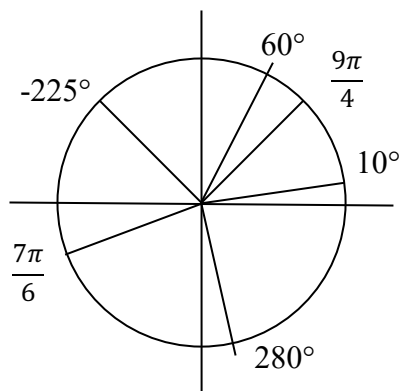


22. $\cot(A) = -\frac{8}{15}$ where $\frac{3\pi}{2} \leq A \leq 2\pi$



Solutions to Exercises

1.



3. $(90^\circ)\left(\frac{\pi}{180^\circ}\right) = \frac{\pi}{2}$

5. $\left(\frac{2\pi}{3}\right)\left(\frac{180^\circ}{\pi}\right) = 120^\circ$

7. 165°

9. $\frac{14\pi}{9}$

11. $645^\circ - 360^\circ = 285^\circ$

13. $-1682^\circ + 5(360^\circ) = 118^\circ$

15. $\left(\frac{23\pi}{9} - 2\pi\right) = \left(\frac{23\pi}{9} - \frac{18\pi}{9}\right) = \frac{5\pi}{9}$

17. $\left(\frac{-7\pi}{2} + 2(2\pi)\right) = \left(\frac{-7\pi}{2} + \frac{8\pi}{2}\right) = \frac{\pi}{2}$

19. $\cos(A) = \frac{12}{13}$, $\tan(A) = \frac{5}{12}$, $\csc(A) = \frac{13}{5}$, $\sec(A) = \frac{13}{12}$, $\cot(A) = \frac{12}{5}$

21. $\sin(A) = -\frac{24}{25}$, $\cos(A) = -\frac{7}{24}$, $\tan(A) = \frac{24}{7}$, $\sec(A) = -\frac{25}{7}$, $\cot(A) = \frac{7}{24}$